Parallel and nested adaptive Stein exponents for superchannel discrimination

Abstract

For every ordered pair of compatible finite-dimensional deterministic superchannels, the parallel and nested adaptive Stein exponents coincide when the type-I error tends to zero after the block-length limit inferior. Accessible input and output memories may have arbitrary finite dimensions; the two superchannels may have different private realization memories. The proof transfers a channel-uniform CP, trace-nonincreasing approximation to one common flagged parallel strategy. It derives an entropy increment bound, applies ordinary-channel amortization, and inducts on the complete nested circuit. The approximation lemma and its uniformity over entangled inputs are established in an appendix.

The equality proved here resolves the nested-versus-parallel conjecture in Hirche's analysis of superchannel discrimination [1, Sec. 5.4], for the Stein exponents and order of limits defined below. Hirche identified a channel asymptotic-equipartition condition sufficient for this conclusion and developed the amortization route [1, Appendix A]. We follow that route and establish the uniform approximation needed here, rather than assume it. Nested strategies use each superchannel as a whole; the result does not settle the advantage of the fully general adaptive strategies considered in [1, Sec. 5.5].

1 Superchannels and the two strategy families

All Hilbert spaces in this paper are nonzero and finite dimensional. Channels are completely positive and trace preserving (CPTP). Two compatible deterministic superchannels have realizations

$$\Theta_i(N) = D_i \circ (N \otimes \mathrm{id}_{S_i}) \circ E_i, \qquad E_i : C \to A \otimes S_i, \qquad D_i : B \otimes S_i \to D, \quad i = 1, 2. \tag{1}$$

The private memories S_1, S_2 need not have the same dimension. For accessible side ports X, Y and a channel $N : A \otimes X \to B \otimes Y$, write $\Theta_i^{X,Y}(N)$ for the same realization with identity action on X before the slot and on Y after it, using the canonical wire permutations. The private memory is never an accessible port.

A parallel n-use strategy consists of arbitrary finite X, Y, one common channel $N_n : A^{\otimes n} \otimes X \to B^{\otimes n} \otimes Y$, one common density operator on $C^{\otimes n} \otimes X$, and one common final binary effect on $D^{\otimes n} \otimes Y$. It inserts N_n into $\Theta_i^{\otimes n}$ under hypothesis i.

The nested family is defined recursively, as whole-channel nesting [1]. A known channel is a circuit of cost zero. A common known deterministic superchannel may wrap any circuit, without changing its number of tested uses. Finally, if an n-use circuit has channel type $A \otimes X \to B \otimes Y$, applying $\Theta_i^{X,Y}$ to that entire channel produces an $(n+1)$-use circuit of type $C \otimes X \to D \otimes Y$. A strategy uses one such circuit, a common density input, and a common binary effect. This recursion specifies the allowed nesting. All known wrappers, states, and effects are fixed independently of the unknown hypothesis. Finite classical records and feedforward may be carried in the accessible wrapper memory. Its finite dimension may depend on the block length. The recursion provides no separate access to the components E_i, D_i or the private memory S_i.

For an acceptance effect Q, let α be rejection under Θ_1 and β acceptance under Θ_2. For $\mathsf{S} \in \{\mathrm{par}, \mathrm{nest}\}$ put

$$\beta_{\varepsilon,n}^{\mathsf{S}} = \inf\{\beta : \text{an } n\text{-use } \mathsf{S} \text{ strategy has } \alpha \le \varepsilon\}, \qquad \zeta_{\mathsf{S}} = \lim_{\varepsilon \downarrow 0} \liminf_{n \to \infty} -\frac{1}{n} \log_2 \beta_{\varepsilon,n}^{\mathsf{S}}. \tag{2}$$

We use $-\log_2 0 = +\infty$. The accept-all effect is available, so $0 \le \beta^{\mathsf{S}}_{\varepsilon,n} \le 1$ and each exponent is in $[0, +\infty]$. Monotonicity in ε gives the outer limit. No optimizer is required.

Theorem 1. *For every pair (1), with the strategy families and order of limits in (2),*

$$\zeta_{\mathrm{nest}}(\Theta_1 \| \Theta_2) = \zeta_{\mathrm{par}}(\Theta_1 \| \Theta_2).$$

2 Ordinary-channel and state estimates

For density operators define the relative entropy in bits by

$$D(\rho\|\sigma) = \operatorname{Tr} \rho(\log_2 \rho - \log_2 \sigma) \quad \text{if } \operatorname{supp} \rho \subseteq \operatorname{supp} \sigma,$$

and $D(\rho\|\sigma) = +\infty$ otherwise. For compatible channels define

$$D_{\mathrm{ch}}(N\|M) = \sup_{T,\xi} D((N \otimes \mathrm{id}_T)(\xi)\|(M \otimes \mathrm{id}_T)(\xi)), \qquad d(N, M) = \sup_{m \ge 1} \frac{1}{m} D_{\mathrm{ch}}(N^{\otimes m}\|M^{\otimes m}), \quad (3)$$

where T ranges over finite reference spaces and ξ over common density inputs. The stabilized reference can be restricted to a copy of the channel input, by purifying the input and using data processing on its purification. The sequence in the regularization is superadditive, so its supremum is also its asymptotic limit. In particular $D_{\mathrm{ch}}(N\|M) \le d(N, M)$. Appending an identity side channel preserves both quantities, by combining references for one inequality and appending a fixed state for the reverse inequality.

The diamond norm here is unhalved:

$$\|T\|_\diamond = \sup_{R,Z:\|Z\|_1 \le 1} \|(T \otimes \mathrm{id}_R)(Z)\|_1.$$

Unsubscripted norms of operators are Hilbert-space operator norms. The notation $L \le_{\mathrm{CP}} cM$ means that $cM - L$ is completely positive.

Lemma 2 (Uniform dominated approximation)**.** *Let N, M be finite-dimensional channels with $d(N, M) < \infty$. For every real $a > d(N, M)$ there exist CP, trace-nonincreasing maps L_m for every positive integer m and numbers $\delta_m \ge 0$ tending to zero such that, for every sufficiently large m,*

$$L_m \le_{\mathrm{CP}} 2^{ma} M^{\otimes m}, \qquad \|L_m - N^{\otimes m}\|_\diamond \le \delta_m. \quad (4)$$

One map L_m works before any input state or entangled reference is chosen.

The stronger exponential version is Corollary 5.2 of Gao–Ji–Liu [3]. Appendix A proves the vanishing-error version needed here, retaining CP domination, trace nonincrease, and every sufficiently large integer block length.

Lemma 3 (Normalized state smoothing)**.** *If $s = D(\rho\|\sigma) < \infty$ for density operators on a fixed finite space and $b > s$, there are density operators ν_m with*

$$\|\nu_m - \rho^{\otimes m}\|_1 \longrightarrow 0, \qquad \nu_m \le 2^{mb} \sigma^{\otimes m} \quad \textit{for every sufficiently large } m. \quad (5)$$

Proof. The finite-state smooth-max estimate [2, Lemma 3.4], with purified-distance radius $1/m$, supplies subnormalized ρ'_m with

$$P(\rho'_m, \rho^{\otimes m}) \le 1/m, \qquad \rho'_m \le 2^{ms + O(\sqrt{m \log m})} \sigma^{\otimes m}.$$

Its block-length requirement is eventually satisfied. Let $t_m = \operatorname{Tr} \rho'_m$. Since the comparator in the purified distance is normalized, its fidelity squared is at most t_m, and $t_m \geq 1 - m^{-2}$. Normalization $\nu_m = \rho'_m / t_m$ increases fidelity with $\rho^{\otimes m}$, hence $\|\nu_m - \rho^{\otimes m}\|_1 \leq 2/m$. The additional logarithmic domination cost $-\log_2 t_m$ is $o(m)$. The margin $b - s > 0$ absorbs both costs for all large m. Choose arbitrary normalized states at the finitely many earlier lengths. □

We also use the finite-dimensional ordinary-channel amortization identity [2, Theorem 3.5 and Corollary 3.7]:

$$d(P, Q) = \sup_{T,\rho,\sigma:\, D(\rho\|\sigma)<\infty} \left[D((P \otimes \mathrm{id}_T)(\rho) \| (Q \otimes \mathrm{id}_T)(\sigma)) - D(\rho\|\sigma) \right]. \tag{6}$$

We apply it with $\operatorname{supp} J_P \subseteq \operatorname{supp} J_Q$, equivalently finite channel max-relative entropy, and verify this support condition at the point of use. Here J_P is the Choi matrix. Finite input relative entropy makes the subtracted quantity finite.

For finite density operators τ, ω, the direct quantum Stein lemma [4] gives, for every real $r < D(\tau\|\omega)$, effects Q_m with null rejection tending to zero and

$$\operatorname{Tr}(Q_m \omega^{\otimes m}) \leq 2^{-mr} \quad \text{for all sufficiently large } m. \tag{7}$$

In the infinite-relative-entropy case this can be seen directly. If Π projects onto $\ker \omega$ and $u = \operatorname{Tr}(\Pi\tau) > 0$, the effect $I - (I - \Pi)^{\otimes m}$ has alternative acceptance zero and null rejection $(1-u)^m$. Thus singular pairs are included.

3 A common flagged parallel strategy

Put $\mathcal{B} = \zeta_{\mathrm{par}}(\Theta_1 \| \Theta_2)$. Fix channels $N, M : A \otimes X \to B \otimes Y$ and density operators ρ, σ on $C \otimes X$. Denote

$$\tau = \Theta_1^{X,Y}(N)(\rho), \qquad \omega = \Theta_2^{X,Y}(M)(\sigma), \qquad d_0 = d(N, M), \qquad s = D(\rho\|\sigma).$$

Lemma 4. *If d_0, s are finite, then, as an extended-real inequality,*

$$D(\tau\|\omega) \leq \mathcal{B} + d_0 + s. \tag{8}$$

Proof. Choose $a > d_0$ and $b > s$ and take L_m, ν_m from Lemmas 2 and 3. For a CP slot T, let

$$\Gamma_{i,m}(T) = (D_i^{\otimes m} \otimes \mathrm{id}_{Y^{\otimes m}}) \circ (T \otimes \mathrm{id}_{S_i^{\otimes m}}) \circ (E_i^{\otimes m} \otimes \mathrm{id}_{X^{\otimes m}}), \tag{9}$$

with the fixed wire permutations. This construction preserves CP order and preserves trace nonincrease. The latter assertion follows because pre- and postprocessing are trace preserving and a CP trace-nonincreasing map remains so after every reference extension.

For each m, the effect $H_m = I - L_m^*(I)$ is positive. Choose any density operator η_m on $B^{\otimes m} \otimes Y^{\otimes m}$ and define $J_m(Z) = \operatorname{Tr}(H_m Z)\eta_m$. This is CP and $L_m + J_m$ is CPTP. Let $F = \mathbb{C}^2$ be a new accessible flag, and use the single common inserted channel

$$\widehat{L}_m(Z) = L_m(\operatorname{Tr}_F Z) \otimes |0\rangle\langle 0|_F + J_m(\operatorname{Tr}_F Z) \otimes |1\rangle\langle 1|_F. \tag{10}$$

It is CPTP on its entire domain, including all reference extensions. Insert it with the same input $\nu_m \otimes |0\rangle\langle 0|_F$ under both hypotheses. The flag travels through the accessible identity side port. Expanding the physical composition gives the normalized output

$$\Omega_{i,m} = \Gamma_{i,m}(L_m)(\nu_m) \otimes |0\rangle\langle 0|_F + \Gamma_{i,m}(J_m)(\nu_m) \otimes |1\rangle\langle 1|_F. \tag{11}$$

This formula also holds when the success branch is correlated with the private memory: the postprocessing acts on each flag block separately.

Choose the state-test effects Q_m for τ, ω at any strict rate $r < D(\tau\|\omega)$. Accept only on the effect $Q_m \otimes |0\rangle\langle 0|_F$. Its actual unconditional probabilities p'_m, q'_m satisfy, for all large m,

$$p'_m \geq \mathrm{Tr}(Q_m \tau^{\otimes m}) - \delta_m - \|\nu_m - \rho^{\otimes m}\|_1, \tag{12}$$

$$q'_m \leq 2^{m(a+b)} \,\mathrm{Tr}(Q_m \omega^{\otimes m}). \tag{13}$$

For (12), first use diamond-norm contraction through the common pre- and postprocessing in (9), with the private memory treated only as a reference for the diamond estimate. Then use trace-norm contraction for the input-state change; $\Gamma_{1,m}(L_m)$ is CP and trace nonincreasing. In both steps an effect has operator norm at most one. For (13), positivity and CP domination give

$$\Gamma_{2,m}(L_m)(\nu_m) \leq 2^{mb} \Gamma_{2,m}(L_m)(\sigma^{\otimes m}) \leq 2^{m(a+b)} \omega^{\otimes m}.$$

The two domination bounds concern the same physical success branch. There is no domination requirement on J_m.

The repair branch is rejected exactly by the final flag effect. Accordingly the probabilities in (12)–(13) are neither conditioned on success nor divided by its probability. All coefficients of the inserted channel, input, and test may be chosen using the two specified hypotheses, but they are common to the two experiments. The ideal $M^{\otimes m}, \sigma^{\otimes m}$ occur only as comparators in the alternative-probability bound.

By (7) the resulting actual parallel strategies have $\alpha'_m = 1 - p'_m \to 0$ and

$$\beta'_m = q'_m \leq 2^{-m(r-a-b)}.$$

At every fixed $0 < \varepsilon < 1$, these tests are feasible at all sufficiently large integer m. Passing to the type-II infimum and then the limits in (2) gives $\mathcal{B} \geq r - a - b$. Let $a \downarrow d_0$, $b \downarrow s$, and let the strict real rates r increase to $D(\tau\|\omega)$. If this output entropy is infinite, the same inequalities hold for arbitrarily large r and force $\mathcal{B} = \infty$. Thus the argument proves the extended inequality (8) in every case. □

4 The channel increment and the nested converse

Suppose first $\mathcal{B} < \infty$. For fixed N, M with $d(N, M) < \infty$, put $P = \Theta_1^{X,Y}(N)$ and $Q = \Theta_2^{X,Y}(M)$. Any additional finite reference T can be absorbed into the accessible ports as $X \otimes T, Y \otimes T$. The same public reassociation is used under both hypotheses. Identity extension leaves $d(N, M)$ unchanged. Lemma 4 therefore gives, for every finite-cost input pair on that reference extension,

$$D((P \otimes \mathrm{id}_T)(\rho)\|(Q \otimes \mathrm{id}_T)(\sigma)) - D(\rho\|\sigma) \leq \mathcal{B} + d(N, M). \tag{14}$$

To check the support hypothesis for (6), take the same maximally entangled input on the domain of P, Q and its copy. Its input cost is zero. Lemma 4 makes the relative entropy between the normalized Choi outputs finite. Hence $\mathrm{supp}\, J_P \subseteq \mathrm{supp}\, J_Q$, and finite dimensions imply $J_P \leq c J_Q$ for a finite c. Applying (6) to (14) proves

$$d(\Theta_1^{X,Y}(N), \Theta_2^{X,Y}(M)) \leq \mathcal{B} + d(N, M). \tag{15}$$

The approximation thresholds may depend on each fixed reference dimension. The exact pointwise inequality is obtained first, with a right-hand side independent of that dimension, and only then is the reference supremum taken.

For a common known deterministic superchannel Ψ, stabilization and relative-entropy data processing give

$$d(\Psi(P), \Psi(Q)) \le d(P, Q). \tag{16}$$

At each tensor block, absorb its common preprocessing memory into the reference and apply its common postprocessing by data processing. Optimize the common input, divide by the block length, and take the regularization to obtain (16).

Structural induction on the nested circuit now proves that its evaluated channels $K_{n,1}, K_{n,2}$ satisfy

$$d(K_{n,1}, K_{n,2}) \le n\mathcal{B}. \tag{17}$$

A known seed has identical channels and cost zero. A known wrapper preserves the bound by (16). A tested use increases it by at most $\mathcal{B}$ by (15). In particular every intermediate channel cost is finite. This induction covers every finite accessible memory dimension in the recursively defined circuit.

For its final common input and effect, let $p = 1 - \alpha$ and $q = \beta$. State and binary-measurement data processing, followed by (17), give

$$D\big((p, 1-p)\|(q, 1-q)\big) \le D_{\mathrm{ch}}(K_{n,1}\|K_{n,2}) \le d(K_{n,1}, K_{n,2}) \le n\mathcal{B}.$$

If $\alpha \le \varepsilon < 1$, the binary expression obeys

$$D\big((p, 1-p)\|(q, 1-q)\big) \ge p \log_2(1/q) - h_2(p) \ge (1-\varepsilon)\log_2(1/\beta) - 1.$$

Finite $n\mathcal{B}$ excludes $q = 0$ when $p > 0$. Hence every feasible nested test obeys

$$\beta \ge 2^{-(n\mathcal{B}+1)/(1-\varepsilon)}, \qquad \beta_{\varepsilon,n}^{\mathrm{nest}} \ge 2^{-(n\mathcal{B}+1)/(1-\varepsilon)}. \tag{18}$$

The second bound passes to the infimum without an optimizer. It implies

$$\limsup_{n\to\infty} -\frac{1}{n}\log_2 \beta_{\varepsilon,n}^{\mathrm{nest}} \le \frac{\mathcal{B}}{1-\varepsilon}.$$

Taking the ordered limits in (2) yields $\zeta_{\mathrm{nest}} \le \mathcal{B}$.

5 Embedding a parallel circuit in the nested family

We give the routing that supplies the reverse inequality. Let the common joint inserted channel be $N : A_1 \cdots A_n X \to B_1 \cdots B_n Y$. Start with the known channel $J_{n+1} = N$. For $j = n, n-1, \ldots, 1$, set

$$X_j = C_{j+1} \cdots C_n A_1 \cdots A_{j-1} X, \qquad Y_j = D_{j+1} \cdots D_n \mathcal{B}_1 \cdots B_{j-1} Y.$$

The empty products at the endpoints are one-dimensional. Known public wire permutations view $J_{j+1,i}$ as a channel $A_j \otimes X_j \to B_j \otimes Y_j$. Apply the whole tested use $\Theta_i^{X_j, Y_j}$ to obtain $J_{j,i}$, with domain $C_j \otimes X_j$ and codomain $D_j \otimes Y_j$. This is exactly the nested recursion and uses n tested superchannels.

Expanding its realization applies the preprocessings in the order $E_1, \ldots, E_n$, then the joint N, then the postprocessings in the order $D_n, \ldots, D_1$. Preprocessings on disjoint external factors commute, as do the postprocessings on their respective output and private memory factors. The expanded circuit consequently equals the parallel tensor realization. Each private S_j remains inside its own tested use; the common permutations act only on accessible external systems. The final common input and effect have exactly the parallel Born probabilities. This routing is the parallel-to-nested inclusion [1, Appendix C] with its side-port dimensions made explicit.

Therefore $\beta_{\varepsilon,n}^{\mathrm{nest}} \le \beta_{\varepsilon,n}^{\mathrm{par}}$ and $\zeta_{\mathrm{nest}} \ge \mathcal{B}$. Together with the converse, this proves Theorem 1 for finite $\mathcal{B}$. If $\mathcal{B} = +\infty$, inclusion already forces the nested exponent to be $+\infty$, proving the extended-real equality as well.

A A uniform CP, trace-nonincreasing approximation

We prove Lemma 2 using the testing-to-dilation and tensor rate-reduction mechanism of [3, Theorem 2.2]. The maps constructed below are chosen in fixed environments and precede all choices of input or reference.

Fix finite Stinespring isometries $V : X_0 \to Y_0 \otimes E$ and $W : X_0 \to Y_0 \otimes F$ of N, M. Let $d = d(N, M) < \infty$. The normalized Choi states are outputs on a common maximally entangled input, so their relative entropy is at most $D_{\mathrm{ch}}(N\|M) \le d$. Thus $\operatorname{supp} J_N \subseteq \operatorname{supp} J_M$ and, for some finite real L, $J_N \le 2^L J_M$. Kraus-column factorization gives an environmental map $C_* : F \to E$ with

$$V = (I \otimes C_*)W, \qquad \|C_*\|^2 \le 2^L. \tag{19}$$

To verify this in the prescribed, possibly nonminimal environments, write $J_N = F_N F_N^\dagger$, $J_M = F_M F_M^\dagger$. The operator inequality permits $F_N = F_M D$ with $\|D\| \le 2^{L/2}$; this follows by the support pseudoinverse or Douglas factorization. In the fixed Kraus bases take $C_* = D^T$. Complex rectangular transpose preserves singular values. This proof allows redundant and zero Kraus columns. Enlarge L freely when needed.

A.1 Weak approximation uniform over inputs

For $t > 0$, let $E_t(N, M)$ be the maximum of $p - tq$ over common entangled inputs and binary effects. With unnormalized Choi matrices, its dual is

$$E_t(N, M) = \min_{Z \ge 0,\ Z \ge J_N - tJ_M} \|\mathrm{Tr}_{Y_0} Z\|_\infty. \tag{20}$$

Here is the finite SDP justification. The primal variables are a density matrix ξ on X_0 and $0 \le T \le I_{Y_0} \otimes \xi$, with objective $\mathrm{Tr}[(J_N - tJ_M)T]$. The dual also has the inequality $\mathrm{Tr}_{Y_0} Z \le \lambda I$ and minimizes λ. The choices $\xi = I/\dim X_0$, $T = I/(2 \dim X_0)$ are strictly feasible. A sufficiently large $Z = cI$, followed by $\lambda > c \dim Y_0$, is strictly dual feasible. Slater duality yields (20). Bounded dual sublevels have $\mathrm{Tr}\, Z \le \lambda \dim X_0$, so positivity gives compactness and attainment. The primal captures arbitrary references by purification and the input-copy reduction stated after (3).

For an attaining Z, put $A_0 = tJ_M$ and $H = A_0 + Z \ge J_N$. With the Moore–Penrose inverse H^+, define $F_0 = A_0 H^+ F_N$ and $F_1 = Z H^+ F_N$. Support inclusion gives $F_0 + F_1 = F_N$, and

$$F_0 F_0^\dagger \le A_0 H^+ A_0 \le A_0, \qquad F_1 F_1^\dagger \le Z H^+ Z \le Z.$$

Factor $F_0 = F_M D$ with $\|D\| \le \sqrt{t}$ and take $C = D^T$. The residual Stinespring operator has Kraus columns F_1, and therefore

$$\|V - (I \otimes C)W\|^2 = \|(\mathrm{Tr}_{Y_0} F_1 F_1^\dagger)^T\|_\infty \le E_t(N, M), \qquad \|C\| \le \sqrt{t}. \tag{21}$$

All inequalities involve the full operators, rather than one chosen input vector.

For tensor powers at $t = 2^{mr}$ and $r > d$, binary data processing bounds the testing advantage by

$$E_{2^{mr}}(N^{\otimes m}, M^{\otimes m}) \le \min\{1, (md + 1)/(mr)\}. \tag{22}$$

Indeed, positive advantage implies $q < 2^{-mr}$. The binary entropy bound $D((p, 1-p)\|(q, 1-q)) \ge p \log_2(1/q) - 1$ gives $p \le (md + 1)/(mr)$, and the advantage is at most p. If $q = 0$, finite channel entropy forces $p = 0$. Combining (21)–(22), for any $\sqrt{d/r} < \delta < 1$, supplies maps C_m^{w} at every large integer m with

$$\|C_m^{\mathrm{w}}\| \le 2^{mr/2}, \qquad \|V^{\otimes m} - (I \otimes C_m^{\mathrm{w}})W^{\otimes m}\| \le \delta. \tag{23}$$

A.2 Finite rate descent with vanishing error

Choose $d < r < a$ and enlarge L so $L > r$. Suppose accurate environmental maps C_m^{a} are available at a rate $S \in [r, L]$, with

$$\|C_m^{\mathrm{a}}\| \le 2^{mS/2}, \qquad e_m := \|V^{\otimes m} - (I \otimes C_m^{\mathrm{a}})W^{\otimes m}\| \le K \exp(-\gamma m^{\alpha}), \quad 0 < \alpha \le 1,$$

for all large m. The exact maps $C_*^{\otimes m}$ supply the initial case $S = L$ with zero error. Let

$$X_m = (I \otimes C_m^{\mathrm{w}})W^{\otimes m}, \qquad Y_m = (I \otimes C_m^{\mathrm{a}})W^{\otimes m}, \qquad H_m = Y_m - X_m.$$

Eventually $\|X_m\| \le A := 1 + \delta$ and $\|H_m\| \le B_0 := (1 + \delta)/2 < 1$. Choose $0 < \beta < 1$ close enough to one that for some $\kappa > 0$

$$h(u) + (1 - u) \ln A + u \ln B_0 \le -\kappa \quad (\beta \le u \le 1),$$

where h is binary entropy with natural logarithms. This is possible by continuity, since the expression at $u = 1$ is $\ln B_0 < 0$.

Expand $Y_m^{\otimes k} = (X_m + H_m)^{\otimes k}$ and retain only words with at most $\lfloor \beta k \rfloor$ factors H_m. The omitted physical operator has norm at most

$$\sum_{j > \beta k} \binom{k}{j} A^{k-j} B_0^j \le (k + 1) e^{-\kappa k}, \tag{24}$$

using $\binom{k}{j} \le e^{kh(j/k)}$. The retained environmental coefficient has norm at most

$$\sum_{j \le \beta k} \binom{k}{j} 2^j 2^{m[(k-j)r + jS]/2} \le 3^k 2^{mk[(1-\beta)r + \beta S]/2}. \tag{25}$$

This follows because H_m is obtained from the coefficient $C_m^{\mathrm{a}} - C_m^{\mathrm{w}}$, of norm at most $2\, 2^{mS/2}$, in the same fixed alternative environment.

For every sufficiently large total integer length M, take $m = \lfloor \sqrt{M} \rfloor$, $k = \lfloor M/m \rfloor$, and $\ell = M - mk < m$. Tensor the retained coefficient with the exact $C_*^{\otimes \ell}$, and reindex tensor factors publicly. Its squared norm has base-two logarithm at most

$$mk[(1 - \beta)r + \beta S] + 2k \log_2 3 + \ell L.$$

Dividing by M makes the last two terms vanish. Thus any fixed $R > (1 - \beta)r + \beta S$ bounds that logarithm by MR at all large integer M. The physical error is at most

$$(k + 1)e^{-\kappa k} + k e_m (1 + e_m)^{k-1}.$$

The second term is the telescoping difference between $Y_m^{\otimes k}$ and $V^{\otimes mk}$. Since m, k are of order $\sqrt{M}$, it and the first term are bounded, after enlarging and decreasing constants, by $K' \exp(-\gamma' M^{\alpha/2})$. Exact remainder padding has operator norm one after action on $W^{\otimes \ell}$ and adds no error. This constructs an accurate family at rate R for every large integer M, not only a subsequence of block lengths.

Let $\theta = (1 + \beta)/2 \in (\beta, 1)$ and set $S_j = r + \theta^j (L - r)$. Each S_{j+1} is strictly above $(1 - \beta)r + \beta S_j$. Starting from the exact family at $S_0 = L$, the preceding step gives accurate families at all S_j. Choose a finite j with $S_j < a$. Although each step halves the stretched-exponential power, after finitely many steps it remains positive. We have therefore constructed C_m before any input or reference choice with

$$\|C_m\| \le 2^{ma/2}, \qquad e_m = \|V^{\otimes m} - (I \otimes C_m)W^{\otimes m}\| \longrightarrow 0. \tag{26}$$

A.3 CP order, trace nonincrease, and the diamond estimate

Put $\widehat{V}_m = (I \otimes C_m)W^{\otimes m}$ and $\widetilde{V}_m = \widehat{V}_m/(1+e_m)$. Since $V^{\otimes m}$ is an isometry and $\|\widehat{V}_m - V^{\otimes m}\| = e_m$, the operator $\widetilde{V}_m$ is a contraction and $\|\widetilde{V}_m - V^{\otimes m}\| \le 2e_m$. Define

$$L_m(Z) = \mathrm{Tr}_{E^{\otimes m}} \widetilde{V}_m Z \widetilde{V}_m^\dagger.$$

This is CP and trace nonincreasing on the entire input algebra. In the fixed Kraus-column matrices, its environmental coefficient is $C_m/(1+e_m)$. Its squared norm is at most 2ma, so $J_{L_m} \le 2^{ma} J_M^{\otimes m}$, which is exactly CP domination.

For every reference R and every complex Z of trace norm at most one, the two-term product difference and contraction of partial trace give

$$\|[(L_m - N^{\otimes m}) \otimes \mathrm{id}_R](Z)\|_1 \le \|\widetilde{V}_m - V^{\otimes m}\|(\|\widetilde{V}_m\| + \|V^{\otimes m}\|)\|Z\|_1 \le 4e_m.$$

All operator norms are unchanged by tensoring with I_R. Thus (4) holds with $\delta_m = 4e_m$ eventually. At the finitely many earlier lengths take $L_m = 0$ and any finite error bound. This supplies CP/TNI maps at every length and proves the lemma.

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